

MANGESH RAJENDRA SAVANT

mangesh.pvt24@gmail.com | +91 7588086319 | [Linkedin](#) | [Leetcode](#) | [Github](#)

AIML Engineer

AI/ML engineering student at VIT Pune (GPA 9.55) focused on computer vision and time-series forecasting. Built production-grade ML systems for national-scale environmental data and real-time AR applications.

EDUCATION

Vishwakarma Institute of Technology, Pune

Aug 2024 – Present

B.Tech – Artificial Intelligence | GPA: 9.55 | **GPA: 9.55**

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Linear Algebra, Probability & Statistics

KRT Arts, BH Commerce & AM Science College, Nashik

Jun 2024

12th HSC Board | **87.67%**

Seth Lalji Narayanji Sarvajanik Vidyalaya, Jalgaon

Jun 2022

10th SSC Board | **93.80%**

PROJECTS

360° Fashion Try-On with AR/VR & Generative AI

2025

- Built a 360° virtual try-on system integrating 3D body pose estimation and generative diffusion models to render photorealistic garment overlays on user avatars from multiple viewing angles.
- Engineered a real-time rendering pipeline combining ControlNet-based pose conditioning and NeRF-style view synthesis to handle occlusion, lighting variation, and fabric deformation.
- Reduced latency for garment overlay inference by 40% through model quantization and GPU-accelerated inference, making it deployable in mobile AR shopping environments.

Technologies: Python, PyTorch, Diffusion Models, ControlNet, 3D Pose Estimation, OpenCV, AR/VR, ONNX

BHUJAL-NETRA – Real-Time Groundwater Evaluation Platform

2025

- Developed a full-stack groundwater monitoring and forecasting platform for 5,260+ DWLR stations across India; reached Local Finals at Smart India Hackathon 2025.
- Implemented multi-model forecasting pipeline using LSTM for short-term water level prediction and Pastas + Water Table Fluctuation model for recharge estimation; LSTM model achieved RMSE of 0.18m over a 30-day prediction horizon.
- Delivered a Flutter mobile app with multilingual AI chatbot, policymaker dashboard, research portal, and GIS-based region-wise groundwater classification map.

Technologies: Python, LSTM, Pastas, Flutter, Firebase, Matplotlib, Google Maps API

OCEP – Object Communication Error Predictor

Jan 2026

- Designed an interactive Streamlit dashboard for uploading research datasets, visualizing software quality metrics, and predicting object communication errors using ML classifiers.
- Implemented ensemble ML prediction pipelines with feature importance visualization, achieving reliable software reliability analysis on benchmark datasets.

Technologies: Python, Streamlit, Scikit-learn, Machine Learning, Data Visualization

Malware Detection System

Jan 2026

- Developed a machine learning-based malware detection system classifying malicious vs. benign software through behavioral and feature-based analysis of executable files.
- Built an end-to-end preprocessing and classification pipeline with feature engineering; achieved 94%+ classification accuracy on benchmark datasets with sub-100ms inference latency.

Technologies: Python, Scikit-learn, Machine Learning, Cybersecurity, Data Analysis

Mark and Sweep Garbage Collection Algorithm

Jan 2025

- Implemented a Mark and Sweep Garbage Collector in Python with object tracking, root set management, and graph-based reachability analysis handling circular references.
- Created visualization and statistical monitoring modules for analysing GC cycles and object lifecycles, simulating real-world memory management techniques.

Technologies: Python, Garbage Collection, Graph Algorithms, OOP, Graphviz

TECHNICAL SKILLS

Domain	Tools & Technologies
AI / ML & Predictive Analytics	Python, Scikit-learn, TensorFlow, LSTM, Pastas, HuggingFace, NumPy, Pandas
Computer Vision & Generative Models	3D Pose Estimation, Diffusion Models, AR/VR Rendering, OpenCV
Data Science & Visualization	Matplotlib, Streamlit, Data Analysis, Statistical Modeling
Full Stack & Cloud	Flutter, Firebase, React, REST APIs, Google Maps API
Programming Languages	Python, C, C++, Java, JavaScript, Go

EXTRA-CURRICULAR ACHIEVEMENTS

Theatre – Multiple Awards (2025)

- Firodiya Karandak 2025: 1st Prize – 3D Miniature
- Sarpotdar Karandak 2025: Best Actor – 3rd Prize
- Vinodottam Karandak 2025: Best Poster Making – 1st Prize
- Crescendo Natyasamrat: 1st Prize
- Active participant in 7+ theatre productions as Actor, Technical Coordinator, Sound Engineer, and Backstage Crew.